

The background is a vibrant green with a gradient from dark at the top to light at the bottom. It features several thin, curved, glowing lines that sweep across the frame, creating a sense of motion and depth. These lines are more prominent on the left side and fade towards the right.

Definition

- Hypertension, or high blood pressure, is defined as a **persistent elevation of arterial blood pressure.**

According to the **World Health Organization (WHO)** and **American Heart Association (AHA)**, it is diagnosed when:

- **Systolic blood pressure (SBP) is 140 mmHg or higher, and/or**
- **Diastolic blood pressure (DBP) is 90 mmHg or higher.**

Classification of Blood Pressure

Category	Systolic (mmHg)	Diastolic (mmHg)
Normal	<120	<80
high normal prehypertension	120–129	<80
Hypertension Stage 1	130–139	80–89
Hypertension Stage 2	≥140	≥90
Hypertensive Crisis	>180	>120

Pathophysiology

- Blood pressure is determined by two main factors:
- **Cardiac output** – the amount of blood the heart pumps.
- **Peripheral vascular resistance** – the resistance of blood vessels to blood flow.
- Hypertension occurs when there is **increased cardiac output, increased vascular resistance**, or both.
Contributing mechanisms include:
- Overactivation of the **sympathetic nervous system**
- Overstimulation of the **renin-angiotensin-aldosterone system (RAAS)**
- **Endothelial dysfunction**
- **Sodium and water retention**

Types of Hypertension

Primary (Essential) Hypertension

- Accounts for **90–95%** of cases.
- No single identifiable cause.
- Risk factors include genetics, age, obesity, stress,

Secondary Hypertension

Has a specific underlying cause, such as:

- Chronic kidney disease
- Endocrine disorders (e.g., Cushing's syndrome, hyperthyroidism, pheochromocytoma)

Risk Factors

- **Non-modifiable:** Age, family history, ethnicity, and gender
- **Modifiable:** Obesity, high salt intake, smoking, alcohol use, physical inactivity, stress, and unhealthy diet

Clinical Manifestations

- Hypertension is often called the “**silent killer**” because many patients are asymptomatic.
However, symptoms can appear when blood pressure is extremely high, including:
- Headache (especially in the morning)
- Dizziness
- Blurred vision
- Chest pain
- Shortness of breath
- Nosebleeds (epistaxis)

Complications

- Untreated or poorly controlled hypertension can lead to **target organ damage**, affecting:
- **Heart:** Left ventricular hypertrophy, heart failure, myocardial infarction
- **Brain:** Stroke, transient ischemic attack
- **Kidneys:** Chronic kidney disease
- **Eyes:** Retinopathy
- **Blood vessels:** Peripheral arterial disease

Diagnosis

- **Accurate blood pressure measurement** using proper technique.
- **Multiple readings** on different days.
- **Laboratory tests** to assess organ function and identify causes:
 - Blood tests: Creatinine, electrolytes, glucose, lipids
 - Urinalysis for protein or albumin
- ECG and sometimes echocardiogram

Management

Lifestyle Modifications (First-line for all patients):

- Reduce salt intake (<5 grams per day)
- Maintain healthy body weight (BMI <25)
- Regular physical activity (at least 30 minutes, 5 times a week)
- Avoid smoking and limit alcohol intake
- Eat a balanced diet rich in fruits, vegetables, and whole grains
- Manage stress through relaxation techniques

Pharmacologic Therapy

If lifestyle changes are not enough, antihypertensive medications are prescribed:

- **Diuretics** (e.g., thiazides)
- **ACE inhibitors**
- **Angiotensin II receptor blockers (ARBs)**
- **Calcium channel blockers**
- **Beta blockers** (in selected cases)

- The goal is to **reduce blood pressure below 140/90 mmHg**, or lower for certain patients such as diabetics or those with kidney disease.

Nursing Management

As nurses, your responsibilities include:

- **Accurate BP measurement** and documentation
- **Monitoring medication compliance and side effects**
- **Educating patients** about lifestyle changes and the importance of adherence
- **Recognizing signs of hypertensive crisis** and acting promptly
- **Coordinating care** with physicians and other

Hypertensive Emergency

- A hypertensive emergency is when **BP exceeds 180/120 mmHg with target organ damage** such as chest pain, confusion, or renal failure. This requires **immediate hospitalization and IV**

Conclusion

- In summary, hypertension is a **chronic, manageable condition** that plays a major role in cardiovascular morbidity and mortality.
Through **early detection, lifestyle modification, and patient education**, nurses can significantly reduce the complications of hypertension and improve

Thank you for listening.